

华南理工大学
South China University of Technology

# Natural Language Processing
# Group Project

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GitHub Repository: https://github.com/lcw578/RAG_project

## Introduction:

MultiHop-RAG (Tang and Yang, 2024) is a benchmark for multi-hop question answering over news: 609 English articles and 2,556 questions, split into inference (816), comparison (856), temporal (583) and null (301). A question names two or more sources, and the facts it needs are spread over separate articles, so a system must retrieve across documents, track provenance, and decide when evidence is insufficient. Null questions are deliberately unanswerable and must be answered "Insufficient Information".

**Research questions.** This project asks three. First, how far can a frozen 9B model go when retrieval is strong but no training is allowed? Second, which layer limits multi-hop accuracy: retrieving the right documents, or reasoning over them? Third, does the question text itself carry supervision-free signals - the source names and publication dates it mentions - that can close part of the gap?

**What the system is.** We built a complete retrieval-augmented generation (RAG) system over the full corpus. The system is RAG by construction: every answer is generated at inference time from evidence retrieved for that question, and answer, evidence_list and question_type are never used as input, index content or labels. Retrieval is the official hybrid pipeline of the benchmark repository; on top of it we added four mechanism-level modules that read only the question and the corpus metadata, a source-aware and time-aware prompt, and an output-validation rule. On all 2,556 questions the official evaluator reports overall Accuracy 0.7700, i.e. benchmark score B = 77.00, which lies in the Excellent band (B >= 75). The reused pipeline alone scores 58.6; the added mechanisms are worth 18.4 points.

## Methodology:

The system is a six-stage pipeline (Table 1, Figure 1). Stages 2-4 run between retrieval and generation and are pure functions of the retrieval result, the question and corpus metadata; this is what makes the Section 3 ablation possible, since every configuration reuses the same base retrieval.

**Table 1. Pipeline stages and key settings.**

| # | Stage | Key setting |
|---|---|---|
| 1 | Hybrid retrieval (reused) | Dense top-10 + BM25 top-10, RRF k = 60, BGE rerank top-10 |
| 2 | Source coverage (R1) | Append 2 chunks per missing named source (<= 3) |
| 3 | Source top-up, date match (R2) | >= 3 chunks/source; +2 per named date; <= 18 total |
| 4 | Temporal annotation (R3) | Prefix each chunk with [Published: YYYY-MM-DD] |
| 5 | Generation (Qwen3.5-9B) | T = 0.1, 512 tokens, thinking off, one call per question |
| 6 | Output validation | Answer of <= 1 character -> "Insufficient Information" |

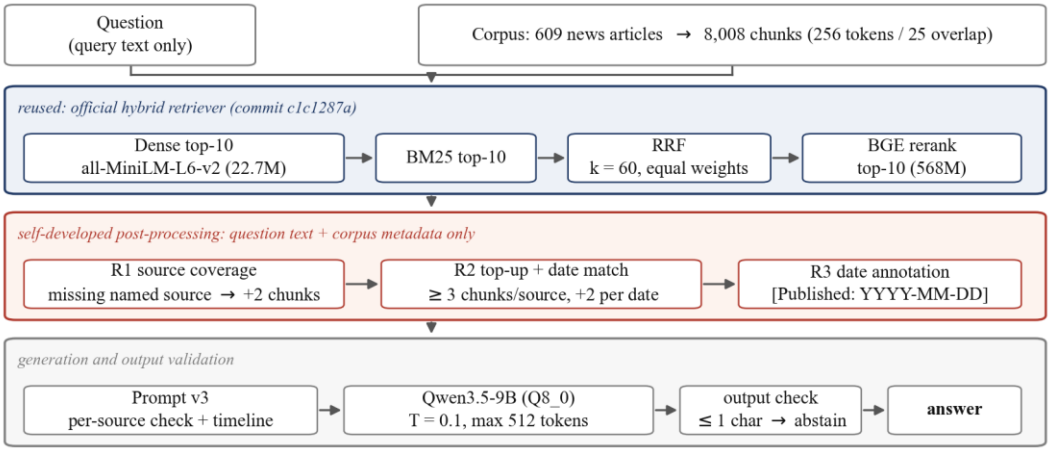



**Figure 1. System data flow. Blue: reused from the official repository (commit c1c1287a). Red: self-developed modules, which read only the question text and corpus metadata.**

**Corpus processing.** The corpus is the complete official set of 609 articles. Documents are split with the official configuration - a sentence splitter with 256-token chunks and 25-token overlap - which yields 8,008 chunks. The overlap is explicit: the splitter default (200) would leave 56 new tokens per chunk and flood the index with offset duplicates. Each chunk keeps its title, source and published_at metadata, and the dense and sparse indexes are built over the chunk text with that metadata prepended, so that source names and dates are searchable rather than invisible to the retriever.

**Retrieval (reused).** Retrieval comes from the official MultiHop-RAG repository (commit c1c1287a); only model paths and the endpoint were adapted, the ranking logic is unchanged. Dense retrieval encodes the question with all-MiniLM-L6-v2 (22.7M parameters) and returns 10 chunks by cosine similarity; sparse retrieval is Okapi BM25 over lower-cased alphanumeric tokens and returns 10 more. The two rankings are merged by reciprocal rank fusion (k = 60, equal weights, duplicates removed by normalised chunk text), and the fused candidates are re-scored by the bge-reranker-v2-m3 cross-encoder (568M parameters); the 10 best chunks form the evidence list and are also the input to the retrieval metrics in Section 3.

**R1 - source coverage guarantee.** Comparison and temporal questions usually name their sources ("as reported by The Verge and TechCrunch"). We match every corpus source name - and its short form before "|" - against the question, with word boundaries and longest name first. When a named source is absent from the 10 retrieved chunks, we score that source's chunks with BM25, take 6 candidates, rerank them and append the best 2. At most 3 sources are completed per question. R1

modified 490 of the 2,556 questions (19.2%).

**R2 - source top-up and date matching.** Presence is not density: a source represented once can still be outvoted by ten chunks from elsewhere. R2 therefore requires at least 3 chunks for every named source (append at most 2 per source, same BM25-then-rerank procedure) and handles explicit dates: when the question names a date such as "October 7, 2023" that exists in the corpus but is absent from the context, it appends 2 chunks published that day. A question receives at most 6 appended chunks and at most 18 chunks in total. Of the 2,019 questions that name at least one corpus source, 693 already met the >= 3 rule, 1,243 were topped up and 83 matched on a date.

**R3 - temporal annotation.** News bodies refer to time relatively ("last Wednesday", "the following year"). Measured on the final contexts, only 1.8% of chunks (487 of 26,586) contain their own publication date as text, so for a temporal question the timeline is not visible to the generator at all. R3 prefixes every chunk with [Published: YYYY-MM-DD] taken from the corpus metadata.

**Generation.** Answers are produced by Qwen3.5-9B, an instruction-tuned model with 9B total parameters (below the 10B limit; quantisation does not change the count), served by llama.cpp (build 11429) as an OpenAI-compatible endpoint. The Q8_0 GGUF file is 9.13 GB. Decoding uses temperature 0.1 and max_tokens 512 with thinking mode disabled; each question is one request over a fresh HTTP connection. The prompt is the official template plus three rules: check what each named source says before answering a cross-source question, and answer Yes only if every named source supports the claim; use the bracketed publication dates when the question asks about order or change; never guess and never answer with a single letter. Prompts average 2,558 tokens (range 1,900-3,922) and contexts average 11.25 chunks.

**Output validation.** An answer of at most one character is replaced by "Insufficient Information". The rule follows the label distribution rather than tuning: no gold answer is a single character, so it cannot damage a correct answer, and it turns degenerate outputs into correct abstentions. In the submitted run it fired 49 times, all on null questions.

**Reuse and own work.** Reused from the official repository (commit c1c1287a): the hybrid retriever, the ingestion and chunking configuration, the generation template and both evaluation scripts, with the scoring logic of qa_evaluate.py untouched. Adaptations are marked [ADAPTED] in the repository and cover only local model paths, the LLM endpoint, the thinking-mode flag and the per-request connection. Self-developed: R1-R3, prompt variants v2 and v3, the output-validation rule, the runners

and the analysis scripts. No third-party RAG framework is used, so every gain in Section 3 can be attributed to a named pipeline stage.

**Compliance.** Only the question text reaches the retriever; R1-R3 read the question plus corpus metadata (source, published_at); the prompt contains the question and the retrieved text. An audit over all 2,556 sent prompts found zero occurrences of any question-type label. Because evidence naturally quotes the answer, we also ran a control experiment: the gold string appears in the prompt for 91.7% (376/410) of questions whose evidence chunk was retrieved, but for only 27.5% (119/432) of questions whose evidence was missed. The rate follows retrieval, which is what co-occurrence looks like, not injection.

**Reproduction.** The six stages run in this order from the repository root:

```
python code/retrieval/run_hybrid_retrieval.py --queries dataset/MultiHopRAG.json \
    --limit 2556 --retrieve_top_k 10 --rrf_top_k 10 --rerank_top_n 10 \
    --output output/full_query_result.json
python code/enhancers/enhance_sources.py && python code/enhancers/enhance_v2.py
python code/enhancers/temporal_annotate.py output/final_query_result_pre.json \
    output/final_query_result.json
python code/generation/qa_llama_v3.py --input output/final_query_result.json \
    --output qa_output/final_v3_full.json
python code/evaluation/qa_evaluate.py --file results/predictions_final.json
```

Artifacts: results/predictions.json (2,556 records, ids q0000-q2555), results/predictions_final.json (the trace of every question: prompt, retrieved evidence with source and date, and answer), results/eval_qa_final.txt and results/eval_retrieval_final.txt (raw evaluator output). Environment: Python 3.11 with requirements.txt, and llama-server -m Qwen_Qwen3.5-9B-Q8_0.gguf --port 18080 -ngl 99 -c 8192 --jinja.

## Evaluation Result:

**Protocol.** All 2,556 questions are scored with the official qa_evaluate.py at commit c1c1287a, unmodified. The script lowercases answers and counts a question as correct when prediction and reference share at least one whitespace-separated token; in this one-answer setting its Precision, Recall, F1 and Accuracy coincide. The benchmark score is B = 100 x overall Accuracy = 77.00. Nothing is dropped: results/predictions.json contains every id from q0000 to q2555 exactly once, and the numbers below come from results/eval_qa_final.txt.

**Table 2. Official accuracy on the complete test set (qa_evaluate.py, c1c1287a).**

| Question type | Questions | Correct | Accuracy |
|---|---|---|---|
| Inference | 816 | 776 | 95.10 |
| Comparison | 856 | 575 | 67.17 |
| Temporal | 583 | 341 | 58.49 |
| Null (unanswerable) | 301 | 276 | 91.69 |
| **Overall** | **2,556** | **1,968** | **77.00** |

B = 77.00 places the Response Accuracy component (40% of the project grade) in the Excellent band, whose component score range is 90-100; the band is fixed by the unrounded overall accuracy, not by any per-type result. Supplementary retrieval metrics from the official retrieval_evaluate.py over the 2,255 questions that have gold evidence: Hits@10 = 0.8395, Hits@4 = 0.7840, MAP@10 = 0.3376, MRR@10 = 0.6756. These describe the reused hybrid retriever on its own top-10; R1-R3 append chunks after rank 10 by design, so they leave these numbers unchanged and act on the generator side instead.

**Ablation.** Table 3 re-runs the full 2,556 questions for every configuration. Rows 1-5 are cumulative; row 6 is the submitted system. Each row changes exactly one mechanism except row 4, which bundles date annotation with the prompt that uses it and is therefore reported as one step.

**Table 3. Ablation on all 2,556 questions, official evaluator (Accuracy in %).**

| Configuration | Comp. | Infer. | Null | Temp. | Overall |
|---|---|---|---|---|---|
| 0 Official hybrid pipeline (thinking off) | 41.0 | 94.0 | 61.8 | 33.4 | 58.6 |
| 1 + answer validation | 41.0 | 94.0 | 90.0 | 33.4 | 62.0 |
| 2 + prompt v2 | 55.1 | 92.0 | 80.7 | 36.5 | 65.7 |
| 3 + source coverage (R1) | 58.4 | 93.1 | 80.1 | 38.8 | 67.6 |
| 4 + temporal annotation + prompt v3 | 64.8 | 94.7 | 75.1 | 57.6 | 73.9 |
| 5 + source top-up and date matching (R2) | 67.2 | 95.1 | 75.4 | 58.5 | 75.1 |
| **6 + answer validation (submitted)** | **67.2** | **95.1** | **91.7** | **58.5** | **77.00** |

**Resource usage (measured).** Hardware: one NVIDIA RTX 4080 SUPER with 32 GB, an Intel Core i5-12400F and 15.8 GB of system RAM, driver 610.88. Generation is one local call per question: 2,556 requests in the submitted run, with no failed request and no retry (the runner supports up to five attempts; the log records "attempt 1" only). Token usage measured with the server's own tokenizer: 6,537,197 prompt tokens (mean 2,558, max 3,922) and 5,053 completion tokens. End-to-end generation of the full set took 23 min 39 s (1,419 s), i.e. 0.55 s per question. The retrieval index is built once and persisted, and reused by every configuration; the six ablation rows add about 15.3k further requests and roughly 2.3 h of GPU time.

## Analysis:

**Where the 18.4 points came from.** Table 3 decomposes the gain into five steps, and two of them dominate. Prompt v2 (+3.7) and the temporal bundle of date annotation with prompt v3 (+6.3) account for 10.0 points, and both act on what the generator sees rather than on what is retrieved. Source coverage (+1.9) and source top-up with date matching (+1.2) add 3.1 points. Answer validation contributes 3.4 points at the start and 1.9 more at the end, in both cases almost entirely on null questions. Comparison rises from 41.0 to 67.2 and temporal from 33.4 to 58.5, while inference stays between 94.0 and 95.1 - what a module set aimed at cross-source and time questions should look like.

**The diagnosis behind each module.** At the baseline, 1,057 questions fail: 505 comparison, 388 temporal, 115 null and 49 inference. Matching gold evidence facts against the retrieved chunks shows where the loss sits. For comparison, 81 failures (16.0%) had no gold fact in context, 269 (53.3%) had some but not all, and 155 (30.7%) had every gold fact - yet 404 of the 505 (80.0%) answered "Insufficient Information". The retriever was delivering the evidence and the generator was refusing to use it, which is exactly what prompt v2 targets (check every named source, never answer with a fragment); comparison then gained 14.1 points. For temporal, 71 failures (18.3%) had no gold fact, 229 (59.0%) partial and 88 (22.7%) complete; the missing ingredient there was the publication date, present in only 1.8% of chunk bodies, hence R3 and prompt v3 (+18.8 points on temporal, of which +131 questions flipped from wrong to right and 21 the other way). Null failures were of a different kind: 88 of 115 (76.5%) were answers of two characters or fewer, which is what the validation rule removes.

**What is left.** After all modules, 588 questions are still wrong: 464 (78.9%) are a wrong yes/no answer, 65 (11.1%) are abstentions on answerable questions and 59 (10.0%) are wrong entities. The yes/no errors are symmetric rather than biased: of 782 questions whose gold answer is "yes", 548 are answered "yes" (70.1%); of 561 whose gold answer is "no", 350 are answered "no" (62.4%). The system does not lean toward one polarity; it fails to decide when two sources frame the same event differently, which a single forward pass handles worst.

**Success case q0000 (inference).** The question asks who the cryptocurrency figure on trial for fraud and conspiracy is, "as reported by both The Verge and TechCrunch". Both named sources are in the context with their dates: The Verge, 2023-09-28, "The trial of Sam Bankman-Fried is likely to be more consequential than just whether the

man himself is found guilty"; TechCrunch, 2023-10-02, "our eyes will be trained on the Manhattan Federal Court for the trial of Sam Bankman-Fried, the disgraced cryptocurrency entrepreneur". The system answers "Sam Bankman-Fried", the gold answer. Two properties matter: the answer is a bare entity with no explanatory clause, which is what token-overlap scoring rewards and what the prompt's "answer directly" rule enforces; and the case is robust to noise, since the remaining chunks are off-topic pieces (CNBC on Coinbase, The Sydney Morning Herald on crypto). Inference questions behave this way 95.1% of the time because the entity is usually stated identically in several sources, so one correct chunk suffices.

**Failure case q0043 (temporal).** "Did the 'Fortune' report on Donald Trump's real estate valuations published on September 26, 2023, disagree with 'The Age' report regarding the allegation that Donald Trump increased the value of his penthouse apartment?" Gold answer: no. System answer: Yes. This is not a retrieval failure: both named sources are present with correct dates - three Fortune chunks from 2023-09-26 and seven Age chunks from 2023-11-07, the last appended by R2. The evidence is explicit and concordant. Fortune reports that "the lawsuit accused Trump and his company of routinely inflating the value of" his properties; The Age reports that "New York State arrived at a valuation that was about a third of Trump's" and that "between 2014 and 2015, Trump increased the value of the apartment to $US327 million from $US200 million". Both articles assert the inflation, so they agree, and the answer is "no". The model evaluates the word "disagree" as "does either report allege wrongdoing?" instead of comparing the two claims' polarity. 202 of the 242 temporal errors and 246 of the 281 comparison errors are wrong yes/no answers of this shape, with the evidence present in the context.

**Null questions and the validation rule.** q0013 asks for "the first letter of the name of the company that ... received the most European patents in 2021 ... mentioned in a Bloomberg article". The benchmark marks it unanswerable, but the question literally asks for a letter, and the model answers "A". The validation rule turns that into "Insufficient Information", which is correct. 49 answers had this form in the submitted run, all on null questions, and they account for the whole difference between 75.08 and 77.00. We report the rule as an output-validation layer, not as reasoning: it recovers a formatting failure, and it is safe here only because no gold answer is a single character (verified on all 2,556 gold answers).

**Retrieval is no longer the bottleneck.** Every gold source listed in evidence_list is represented in the context for 2,486 of 2,556 questions (97.3%). Context size is not monotonically helpful either: at the submitted configuration, questions with 10 chunks

score 74.5 (n = 1,285), those with 11-13 chunks 77.3 (n = 1,063) and those with 14 or more 67.3 (n = 208). Appending chunks beyond the first ten buys coverage without buying accuracy, and the remaining 23 points are a reasoning problem rather than a recall problem.

**Negative results.** Two design choices were rejected on evidence. Thinking mode: on 300 questions with max_tokens 2,048, enabling the reasoning trace drops overall accuracy from 59.0 to 14.3, because 70% of the outputs exhaust the budget before emitting an answer (30 s per question against 0.56 s). Connection reuse: reusing an HTTP connection to llama-server makes every second request return 404 (10/10 reproducible, OK/404 alternating), so each question opens a new connection.

**Limitations.** First, the benchmark has no held-out split: every decision was taken while looking at the same 2,556 questions, so the 18.4-point gain is an upper bound, not an unbiased estimate. Two checks bound the risk rather than removing it: a control group of 2,066 questions untouched by R1 moves by -0.2 points, which also fixes the single-run sampling noise at about 0.2 points, and the prompt-v2 gain replicates on both halves of the data (+7.5 and +6.6). Second, the system is single-pass: no query decomposition, no second retrieval round and no verifier, so a question needing a second hop is answered in one shot. Third, the abstention rule is a length heuristic that works because degenerate answers are short; it would not transfer to a benchmark whose answers may be one character. Fourth, 9B parameters was a hardware constraint; the remaining yes/no comparisons are where a stronger generator would help most.

## Contributions:

**Li Chenwei** (implementation, experiments, analysis): adapted the official retrieval stack to the local models, implemented R1-R3, prompt v2/v3 and the output-validation rule, built the chunked and resume-capable runners, ran the seven-configuration ablation and the robustness checks (control group, split-half, thinking A/B), performed the failure attribution and the compliance audit, and organised the GitHub repository.

**Yang Zhiyuan** (verification, statistics, writing): converted the traces into the required predictions.json (q0000-q2555, id validation and duplicate check), verified consistency between traces and predictions, measured the resource statistics in Section 3 (hardware, token counts, runtime, request counts), and organised and proofread the report.

## Conclusion:

This project delivers a complete RAG system for MultiHop-RAG: the full corpus (609 documents, 8,008 chunks), the full question set (2,556), one locally served 9B model, and an official score of 77.00 in the Excellent band, against 58.6 for the reused hybrid pipeline. The main finding is that the bottleneck moved while the project ran. At the baseline, 80% of comparison failures were the generator refusing to answer with the evidence in front of it; at the end, 79% of residual errors are yes/no polarity decisions where the evidence is present and the comparison is made incorrectly. Retrieval-side mechanisms were cheap and effective - source completion and date matching are worth 3.1 points, and date annotation unlocks 6.3 more together with the time-aware prompt - but retrieval has reached its ceiling, since 97.3% of questions already contain every gold source. The next steps follow from that diagnosis: decompose comparison questions into per-source sub-claims and aggregate the verdicts; re-read only the sentences that mention the compared entities in a second pass; and replace the length heuristic with a calibrated abstention decision. A held-out split or cross-validation would turn this upper bound into a defensible generalisation number.